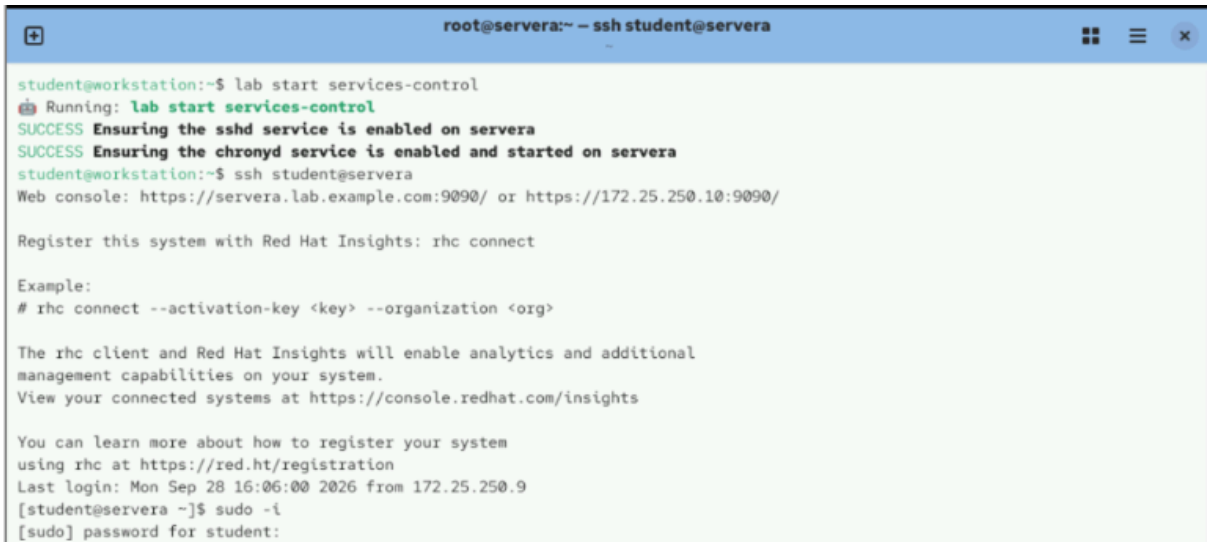
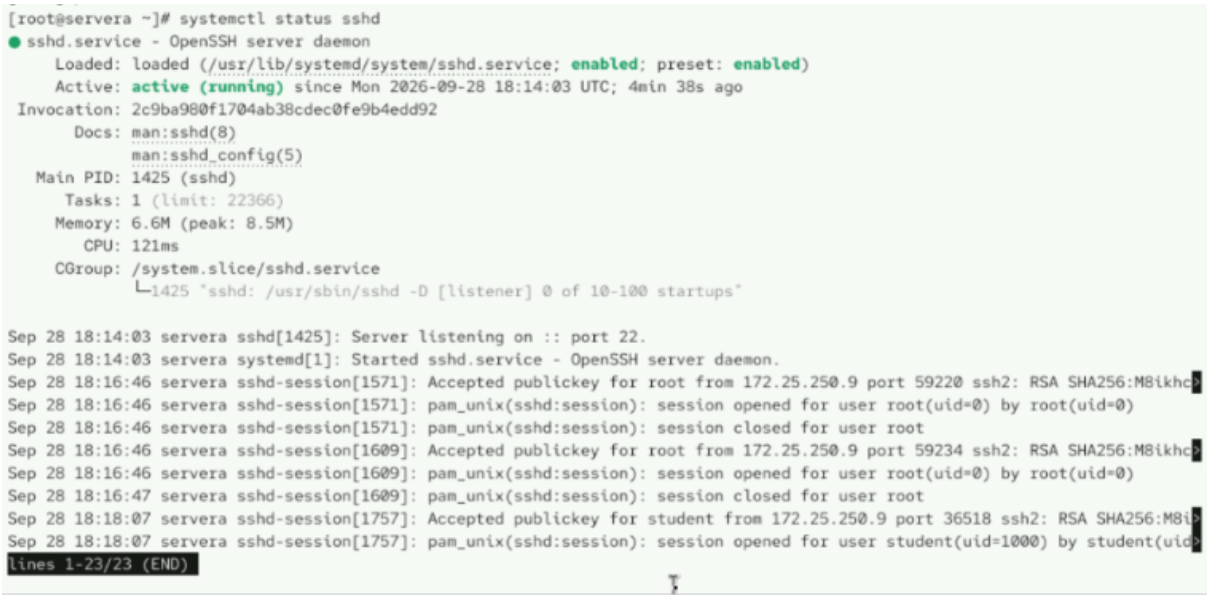
16.4 - Guided Exercise: Control System Services

Ahmed Abdelhedi

1/





```
[root@servera ~]# systemctl restart sshd
[root@servera ~]# systemctl status sshd
● sshd.service - OpenSSH server daemon
     Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-09-28 18:20:09 UTC; 7s ago
 Invocation: cd2e56ed70534105b02693dd307ec91a
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 1938 (sshd)
      Tasks: 1 (limit: 22366)
     Memory: 1M (peak: 1.3M)
        CPU: 14ms
     CGroup: /system.slice/sshd.service
             └─1938 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Sep 28 18:20:09 servera systemd[1]: Starting sshd.service - OpenSSH server daemon...
Sep 28 18:20:09 servera (sshd)[1938]: sshd.service: Referenced but unset environment variable evaluates to an empty string: >
Sep 28 18:20:09 servera sshd[1938]: Server listening on 0.0.0.0 port 22.
Sep 28 18:20:09 servera sshd[1938]: Server listening on :: port 22.
Sep 28 18:20:09 servera systemd[1]: Started sshd.service - OpenSSH server daemon.
lines 1-18/18 (END)
```

```
[root@servera ~]# systemctl reload sshd
[root@servera ~]# systemctl status sshd
● sshd.service - OpenSSH server daemon
     Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-09-28 18:20:09 UTC; 42s ago
 Invocation: cd2e56ed70534105b02693dd307ec91a
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 1963 ExecReload=/bin/kill -HUP $MAINPID (code=exited, status=0/SUCCESS)
   Main PID: 1938 (sshd)
      Tasks: 1 (limit: 22366)
     Memory: 1M (peak: 2.1M)
        CPU: 29ms
     CGroup: /system.slice/sshd.service
             └─1938 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Sep 28 18:20:09 servera systemd[1]: Starting sshd.service - OpenSSH server daemon...
Sep 28 18:20:09 servera (sshd)[1938]: sshd.service: Referenced but unset environment variable evaluates to an empty string: >
Sep 28 18:20:09 servera sshd[1938]: Server listening on 0.0.0.0 port 22.
Sep 28 18:20:09 servera sshd[1938]: Server listening on :: port 22.
Sep 28 18:20:09 servera systemd[1]: Started sshd.service - OpenSSH server daemon.
Sep 28 18:20:44 servera systemd[1]: Reloading sshd.service - OpenSSH server daemon...
Sep 28 18:20:45 servera sshd[1938]: Received SIGHUP; restarting.
Sep 28 18:20:45 servera sshd[1938]: Server listening on 0.0.0.0 port 22.
Sep 28 18:20:45 servera sshd[1938]: Server listening on :: port 22.
Sep 28 18:20:45 servera systemd[1]: Reloaded sshd.service - OpenSSH server daemon.
lines 1-24/24 (END)
```

2/

```
[root@servera ~]# systemctl status chronyd
● chronyd.service - NTP client/server
     Loaded: loaded (/usr/lib/systemd/system/chronyd.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-09-28 18:13:01 UTC; 8min ago
 Invocation: e1ae967ec5ab46eb960f51b62052790e
       Docs: man:chronyd(8)
             man:chrony.conf(5)
   Main PID: 972 (chronyd)
      Tasks: 1 (limit: 22366)
     Memory: 4.4M (peak: 5.1M)
        CPU: 52ms
     CGroup: /system.slice/chronyd.service
             └─972 /usr/sbin/chronyd -F 2

Sep 28 18:13:01 servera systemd[1]: Starting chronyd.service - NTP client/server...
Sep 28 18:13:01 servera chronyd[972]: chronyd version 4.6.1 starting (+CMDMON +NTP +REFCLOCK +RTC +PRIVDROP +SCFILTER +SIGND>
Sep 28 18:13:01 servera chronyd[972]: commandkey directive is no longer supported
Sep 28 18:13:01 servera chronyd[972]: generatecommandkey directive is no longer supported
Sep 28 18:13:01 servera chronyd[972]: Could not open keyfile /etc/chrony.keys
Sep 28 18:13:01 servera chronyd[972]: Frequency -4.905 +/- 2.311 ppm read from /var/lib/chrony/drift
Sep 28 18:13:01 servera chronyd[972]: Loaded seccomp filter (level 2)
Sep 28 18:13:01 servera systemd[1]: Started chronyd.service - NTP client/server.
Sep 28 18:13:07 servera chronyd[972]: Selected source 172.25.254.254
lines 1-22/22 (END)
```

```
[root@servera ~]# systemctl stop chronyd
[root@servera ~]# systemctl status chronyd
○ chronyd.service - NTP client/server
     Loaded: loaded (/usr/lib/systemd/system/chronyd.service; enabled; preset: enabled)
     Active: inactive (dead) since Mon 2026-09-28 18:22:43 UTC; 7s ago
   Duration: 9min 42.671s
 Invocation: e1ae967ec5ab46eb960f51b62052790e
       Docs: man:chronyd(8)
             man:chrony.conf(5)
   Main PID: 972 (code=exited, status=0/SUCCESS)
   Mem peak: 5.1M
        CPU: 54ms

Sep 28 18:13:01 servera chronyd[972]: generatecommandkey directive is no longer supported
Sep 28 18:13:01 servera chronyd[972]: Could not open keyfile /etc/chrony.keys
Sep 28 18:13:01 servera chronyd[972]: Frequency -4.905 +/- 2.311 ppm read from /var/lib/chrony/drift
Sep 28 18:13:01 servera chronyd[972]: Loaded seccomp filter (level 2)
Sep 28 18:13:01 servera systemd[1]: Started chronyd.service - NTP client/server.
Sep 28 18:13:07 servera chronyd[972]: Selected source 172.25.254.254
Sep 28 18:22:43 servera chronyd[972]: chronyd exiting
Sep 28 18:22:43 servera systemd[1]: Stopping chronyd.service - NTP client/server...
Sep 28 18:22:43 servera systemd[1]: chronyd.service: Deactivated successfully.
Sep 28 18:22:43 servera systemd[1]: Stopped chronyd.service - NTP client/server.
[root@servera ~]#
```

3/

```
[root@servera ~]# systemctl is-enabled chronyd
enabled
[root@servera ~]#
```

4/

```
student@workstation:~$ ssh student@servera
Web console: https://servera.lab.example.com:9090/ or https://172.25.250.10:9090/

Register this system with Red Hat Insights: rhc connect

Example:
# rhc connect --activation-key <key> --organization <org>

The rhc client and Red Hat Insights will enable analytics and additional
management capabilities on your system.
View your connected systems at https://console.redhat.com/insights

You can learn more about how to register your system
using rhc at https://red.ht/registration
Last login: Mon Sep 28 18:28:19 2026 from 172.25.250.9
[student@servera ~]$ sudo -i
[sudo] password for student:
```

```
[root@servera ~]# systemctl status chronyd
● chronyd.service - NTP client/server
     Loaded: loaded (/usr/lib/systemd/system/chronyd.service; enabled; preset: ena>
     Active: active (running) since Mon 2026-09-28 18:30:08 UTC; 1min 43s ago
 Invocation: 62bc991fe2f64f8197c12ed8c08145b7
       Docs: man:chronyd(8)
             man:chrony.conf(5)
    Process: 937 ExecStart=/usr/sbin/chronyd $OPTIONS (code=exited, status=0/SUCCE>
   Main PID: 955 (chronyd)
      Tasks: 1 (limit: 22366)
     Memory: 4.4M (peak: 5.2M)
        CPU: 48ms
     CGroup: /system.slice/chronyd.service
             └─955 /usr/sbin/chronyd -F 2

Sep 28 18:30:08 servera systemd[1]: Starting chronyd.service - NTP client/server...
Sep 28 18:30:08 servera chronyd[955]: chronyd version 4.6.1 starting (+CMDMON +NTP>
Sep 28 18:30:08 servera chronyd[955]: commandkey directive is no longer supported
Sep 28 18:30:08 servera chronyd[955]: generatecommandkey directive is no longer su>
Sep 28 18:30:08 servera chronyd[955]: Could not open keyfile /etc/chrony.keys
Sep 28 18:30:08 servera chronyd[955]: Frequency -36.360 +/- 0.956 ppm read from /v>
Sep 28 18:30:08 servera chronyd[955]: Loaded seccomp filter (level 2)
Sep 28 18:30:08 servera systemd[1]: Started chronyd.service - NTP client/server.
Sep 28 18:30:14 servera chronyd[955]: Selected source 172.25.254.254
lines 1-23/23 (END)
```

5/

```
[root@servera ~]# systemctl disable chronyd
Removed '/etc/systemd/system/multi-user.target.wants/chronyd.service'.
[root@servera ~]# systemctl status chronyd
● chronyd.service - NTP client/server
     Loaded: loaded (/usr/lib/systemd/system/chronyd.service; disabled; preset: en>
     Active: active (running) since Mon 2026-09-28 18:30:08 UTC; 2min 35s ago
 Invocation: 62bc991fe2f64f8197c12ed8c08145b7
       Docs: man:chronyd(8)
             man:chrony.conf(5)
   Main PID: 955 (chronyd)
      Tasks: 1 (limit: 22366)
     Memory: 4.4M (peak: 5.2M)
        CPU: 48ms
     CGroup: /system.slice/chronyd.service
             └─955 /usr/sbin/chronyd -F 2

Sep 28 18:30:08 servera systemd[1]: Starting chronyd.service - NTP client/server...
Sep 28 18:30:08 servera chronyd[955]: chronyd version 4.6.1 starting (+CMDMON +NTP>
Sep 28 18:30:08 servera chronyd[955]: commandkey directive is no longer supported
Sep 28 18:30:08 servera chronyd[955]: generatecommandkey directive is no longer su>
Sep 28 18:30:08 servera chronyd[955]: Could not open keyfile /etc/chrony.keys
Sep 28 18:30:08 servera chronyd[955]: Frequency -36.360 +/- 0.956 ppm read from /v>
Sep 28 18:30:08 servera chronyd[955]: Loaded seccomp filter (level 2)
Sep 28 18:30:08 servera systemd[1]: Started chronyd.service - NTP client/server.
Sep 28 18:30:14 servera chronyd[955]: Selected source 172.25.254.254
lines 1-22/22 (END)
```

6/

```
[root@servera ~]# systemctl reboot

Broadcast message from root@servera on pts/1 (Mon 2026-09-28 18:33:03 UTC):

The system will reboot now!

[root@servera ~]# Read from remote host servera: Connection reset by peer
Connection to servera closed.
client_loop: send disconnect: Broken pipe
```

```
student@workstation:~$ ssh student@servera
Web console: https://servera.lab.example.com:9090/ or https://172.25.250.10:9090/

Register this system with Red Hat Insights: rhc connect

Example:
# rhc connect --activation-key <key> --organization <org>

The rhc client and Red Hat Insights will enable analytics and additional
management capabilities on your system.
View your connected systems at https://console.redhat.com/insights

You can learn more about how to register your system
using rhc at https://red.ht/registration
Last login: Mon Sep 28 18:31:25 2026 from 172.25.250.9
[student@servera ~]$
```

```
[student@servera ~]$ systemctl status chronyd
○ chronyd.service - NTP client/server
     Loaded: loaded (/usr/lib/systemd/system/chronyd.service; disabled; preset: en>
     Active: inactive (dead)
       Docs: man:chronyd(8)
             man:chrony.conf(5)
lines 1-5/5 (END)
```

7/

```
[student@servera ~]$ exit
logout
Connection to servera closed.
student@workstation:~$ lab finish services-control
🤖 Running: lab finish services-control
SUCCESS Ensuring the sshd service is enabled on servera
SUCCESS Ensuring the chronyd service is enabled and started on servera
student@workstation:~$
```